

Azure Readiness Day
für HTLs

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Martin Pöckl, Technical Solution Professional

Wien, 6.5.2019

1 | 10:00 Begrüßung

2 | 10:00 – 13:00 Session Teil 1

- Azure Fundamentals (Compute, Storage, Networking)
- Einführung ins Azure Portal
- Development & Test Umgebungen für Studenten

3 | 13:00 – 14:00 *Mittagspause*

7 | 14:00 – 17:00 Session Teil 2

- Azure Data Services
- Cognitive Services
- Datenbanken

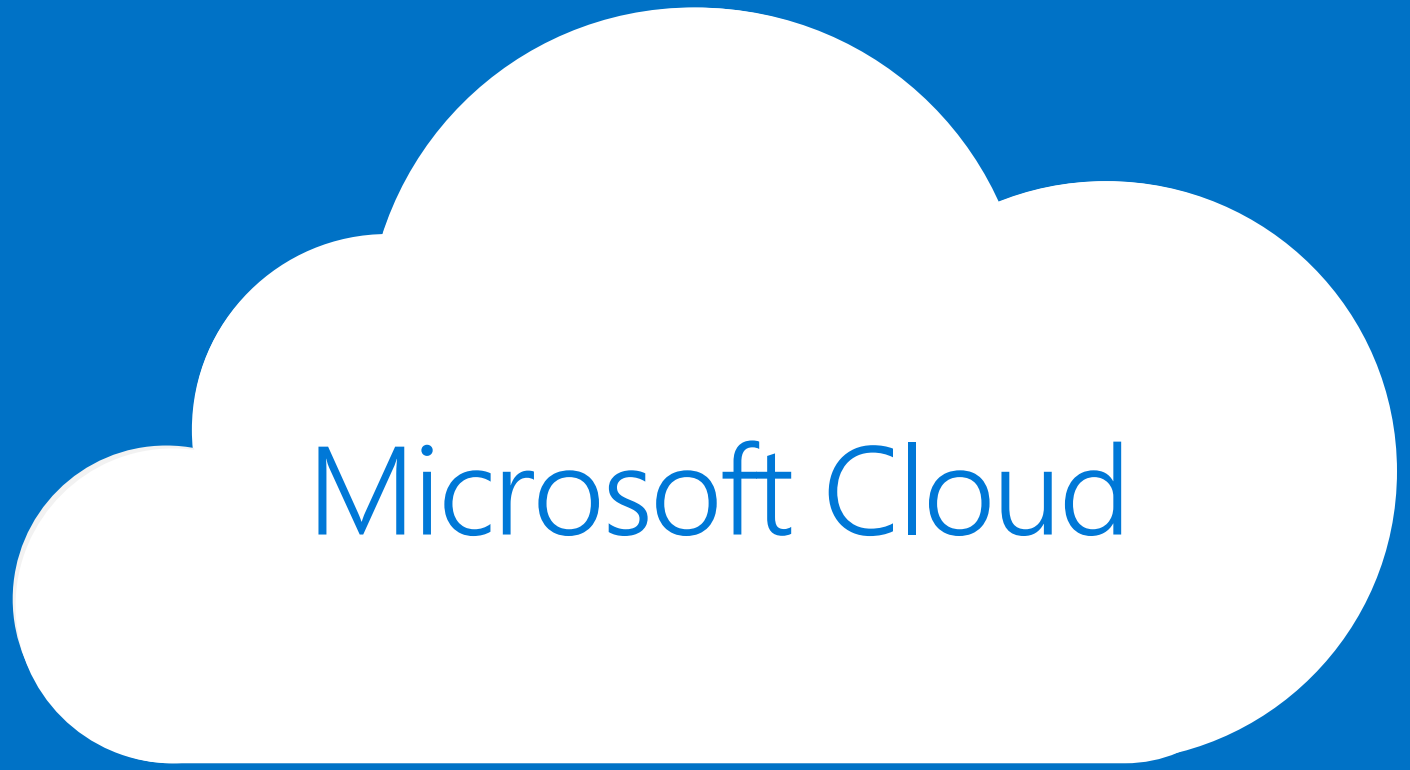


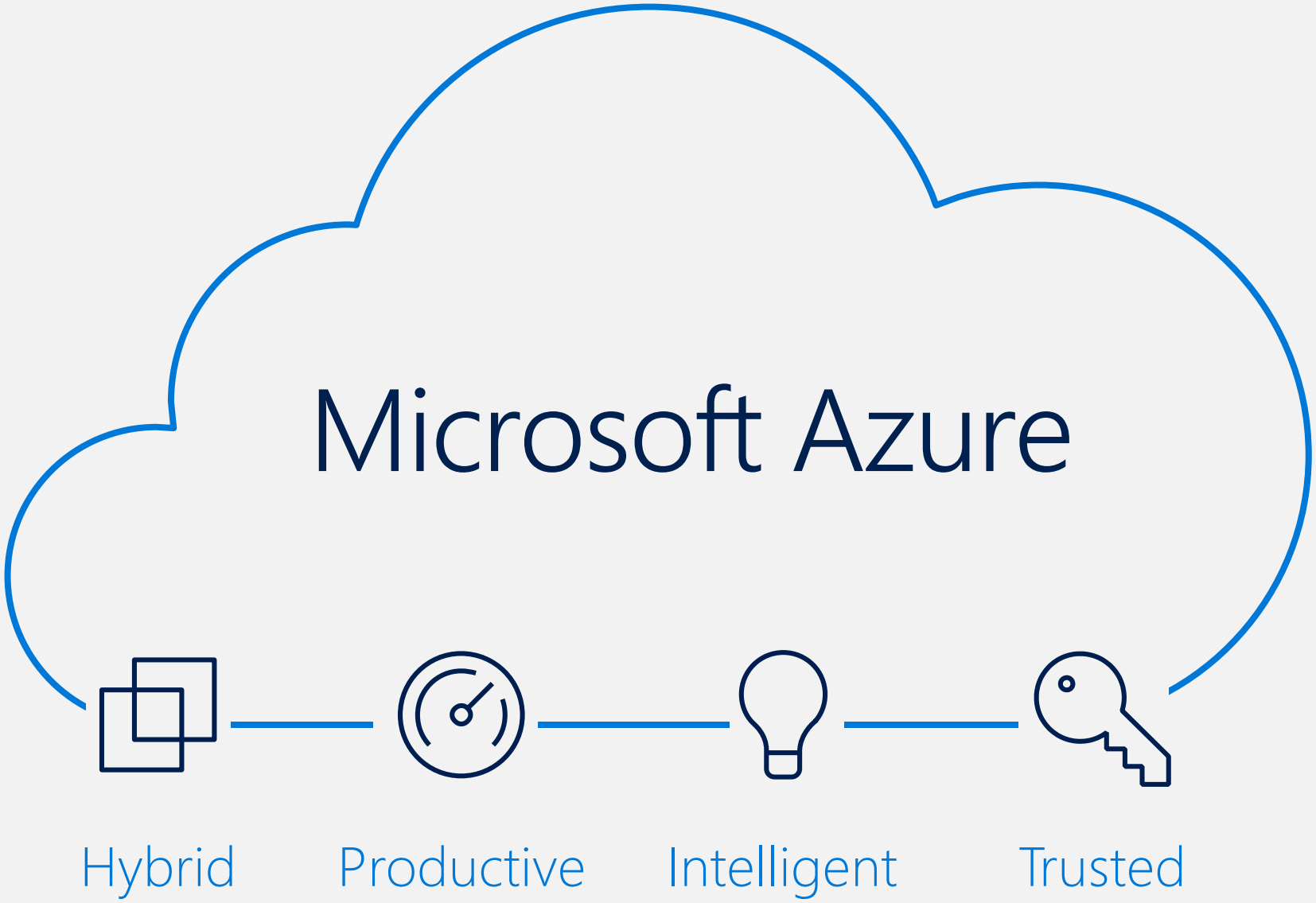
Azure

Dynamics 365

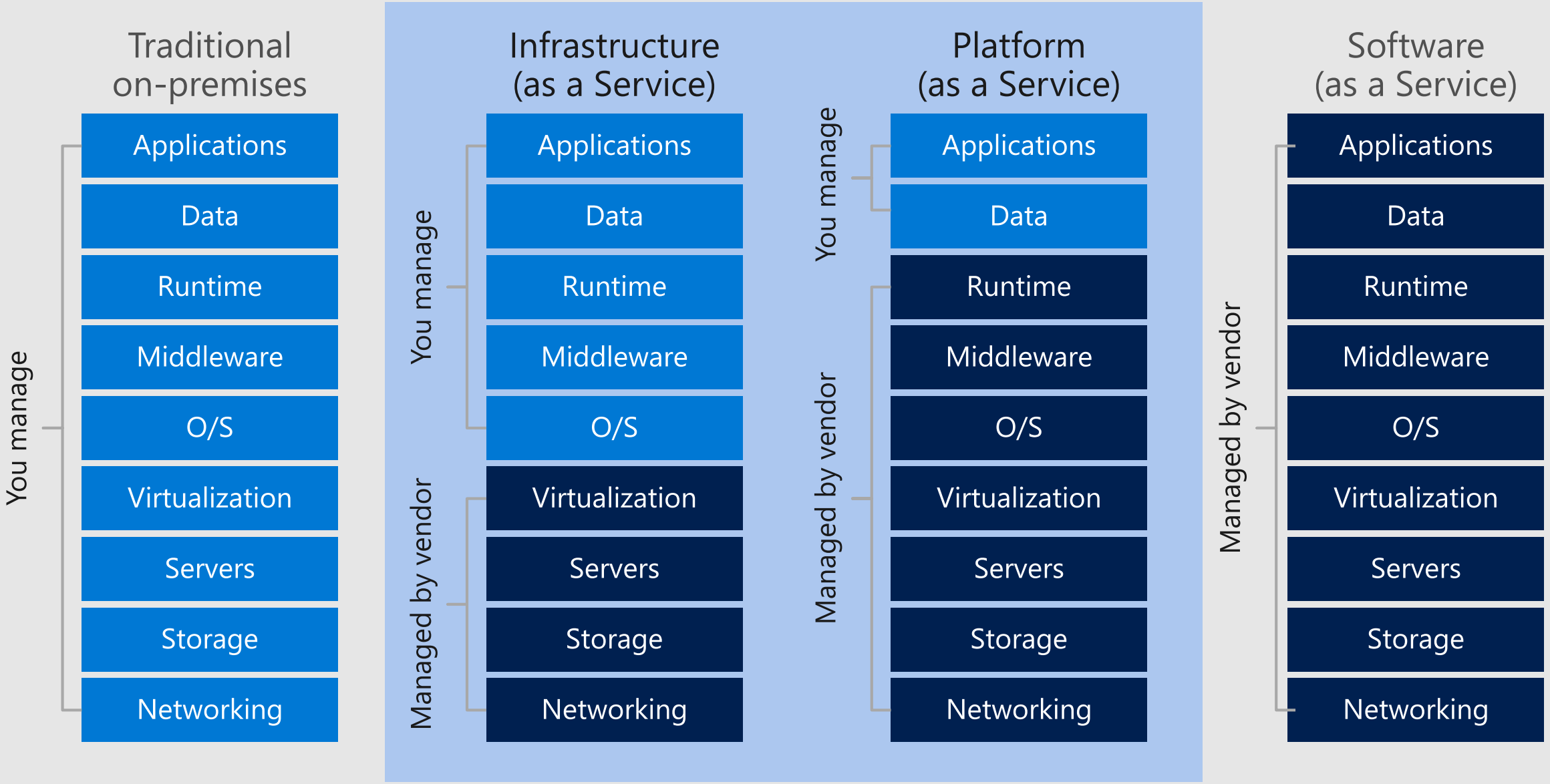
Office 365

Windows


Microsoft Cloud

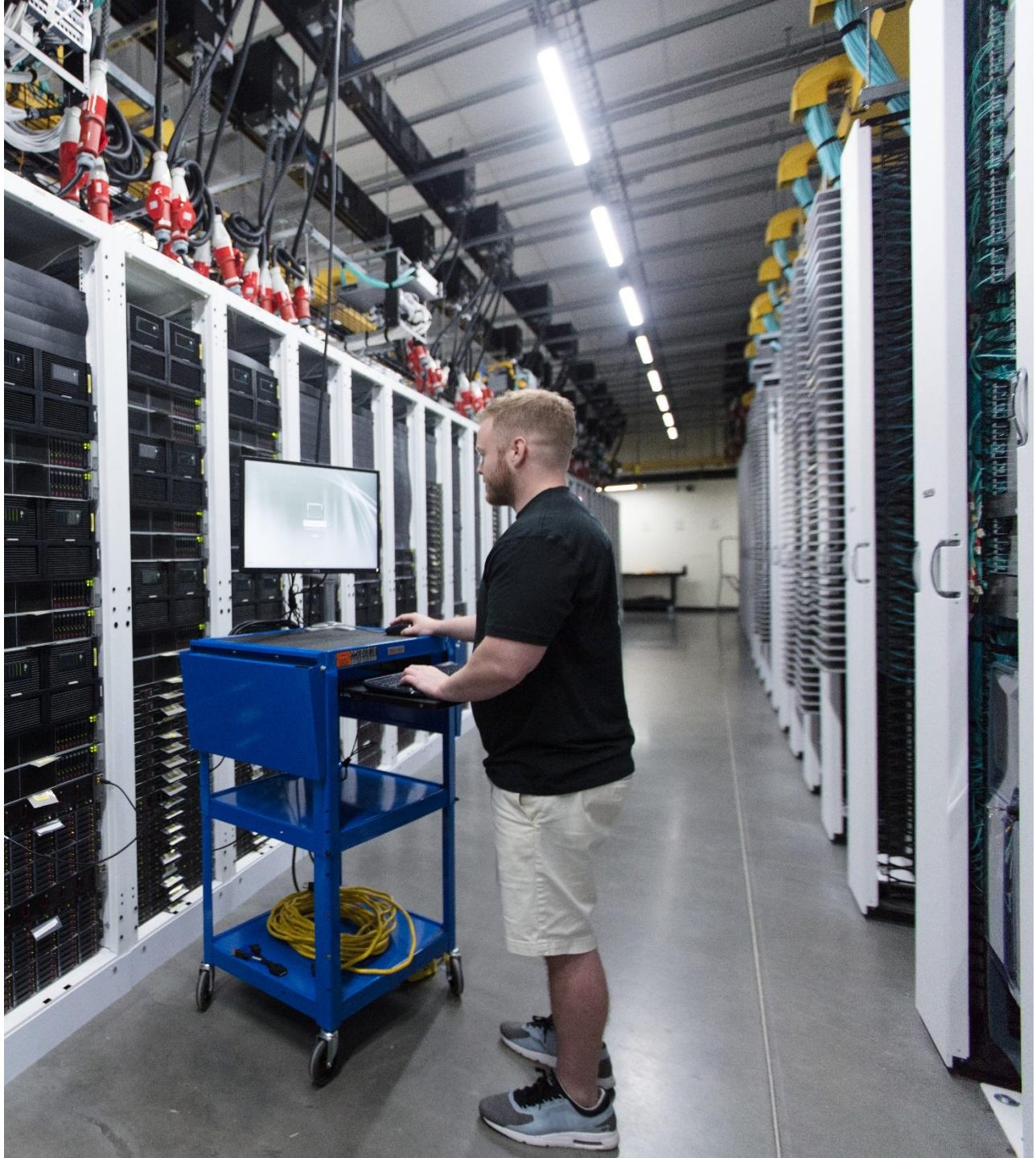

Microsoft Azure
Hybrid
Productive
Intelligent
Trusted

Azure cloud service models



Challenges in maintaining your own infrastructure

- Cost, time and complexity
- Lack of flexibility
- Data explosion
- Efficient energy use



Journey to the Cloud

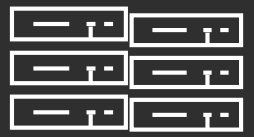
Migrate | **Application Modernization** | **SaaS**

Rehost | Refactor | Rearchitect | Rebuild | Replace



On-Premises

Infrastructure Platform



IaaS

Infrastructure Platform
"lift & shift"



Containers

Managed Container Platform



PaaS/Serverless

Application Platform



SaaS

Microsoft Azure

TOOLS

Visual Studio + Visual Studio Code + Visual Studio Team Services (DevOps)

ADVANCED WORKLOADS

Web + Mobile

Microservices & Container

Serverless

Workflow & Integration

API Management

Identity Management

Data + Analytics

Cognitive Services

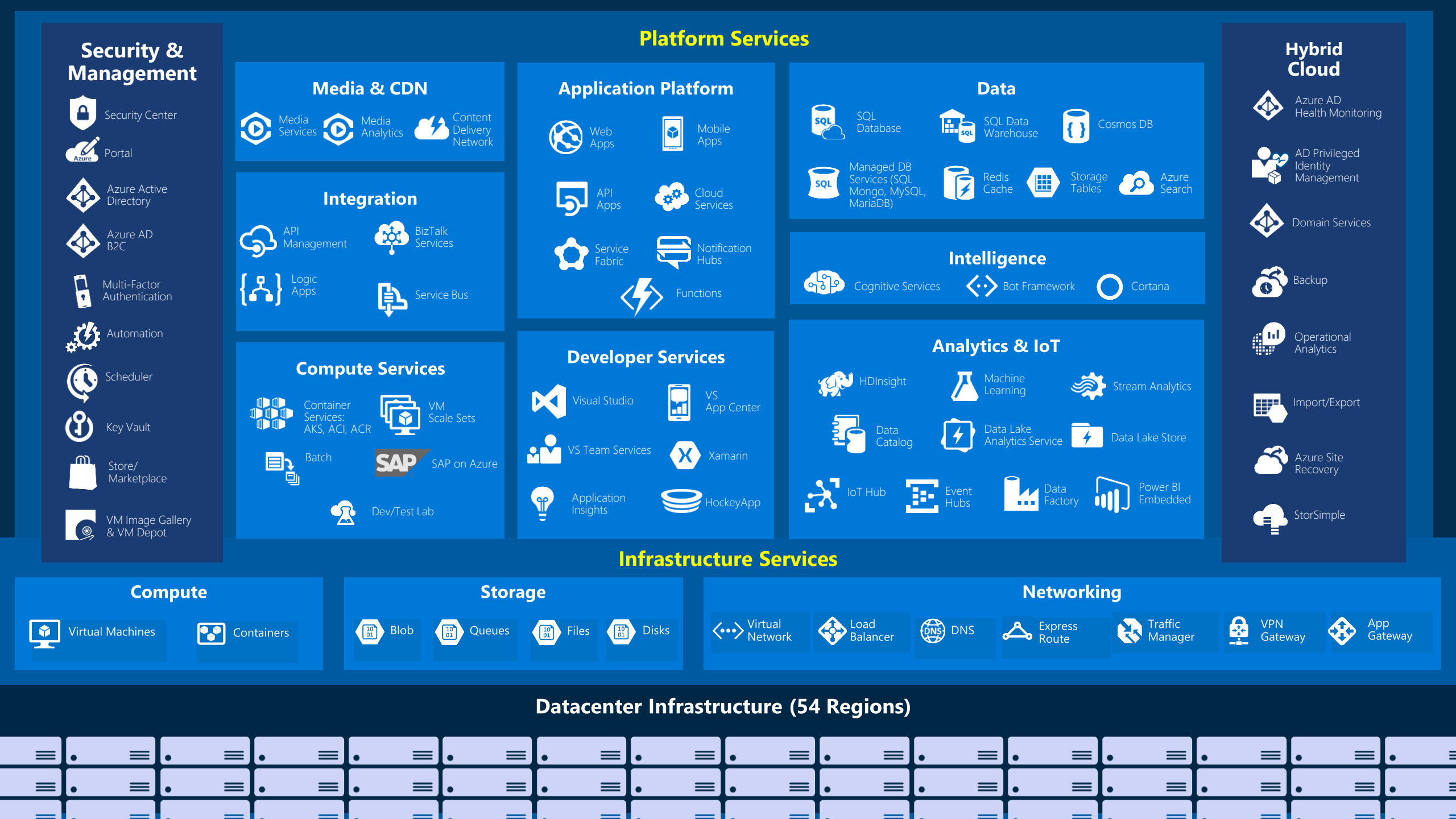
High Performance Compute

Internet of Things

CORE INFRASTRUCTURE

Compute Storage Networking Security

Azure Stack + Hybrid


Security & Management
Security Center
Portal
Azure Active Directory
Azure AD B2C
Multi-Factor Authentication
Automation
Scheduler
Key Vault
Store/ Marketplace
VM Image Gallery & VM Depot
Platform Services
Media & CDN
Media Services
Media Analytics
Content Delivery Network
Integration
API Management
BizTalk Services
Logic Apps
Service Bus
Compute Services
Container Services: AKS, ACI, ACR
VM Scale Sets
Batch
SAP on Azure
Dev/Test Lab
Application Platform
Web Apps
Mobile Apps
API Apps
Cloud Services
Service Fabric
Notification Hubs
Functions
Developer Services
Visual Studio
VS App Center
VS Team Services
Xamarin
Application Insights
HockeyApp
Data
SQL Database
SQL Data Warehouse
Cosmos DB
Managed DB Services (SQL Mongo, MySQL, MariaDB)
Redis Cache
Storage Tables
Azure Search
Intelligence
Cognitive Services
Bot Framework
Cortana
Analytics & IoT
HDInsight
Machine Learning
Stream Analytics
Data Catalog
Data Lake Analytics Service
Data Lake Store
IoT Hub
Event Hubs
Data Factory
Power BI Embedded
Hybrid Cloud
Azure AD Health Monitoring
AD Privileged Identity Management
Domain Services
Backup
Operational Analytics
Import/Export
Azure Site Recovery
StorSimple
Infrastructure Services
Compute
Virtual Machines
Containers
Storage
Blob
Queues
Files
Disks
Networking
Virtual Network
Load Balancer
DNS
Express Route
Traffic Manager
VPN Gateway
App Gateway
Datacenter Infrastructure (54 Regions)

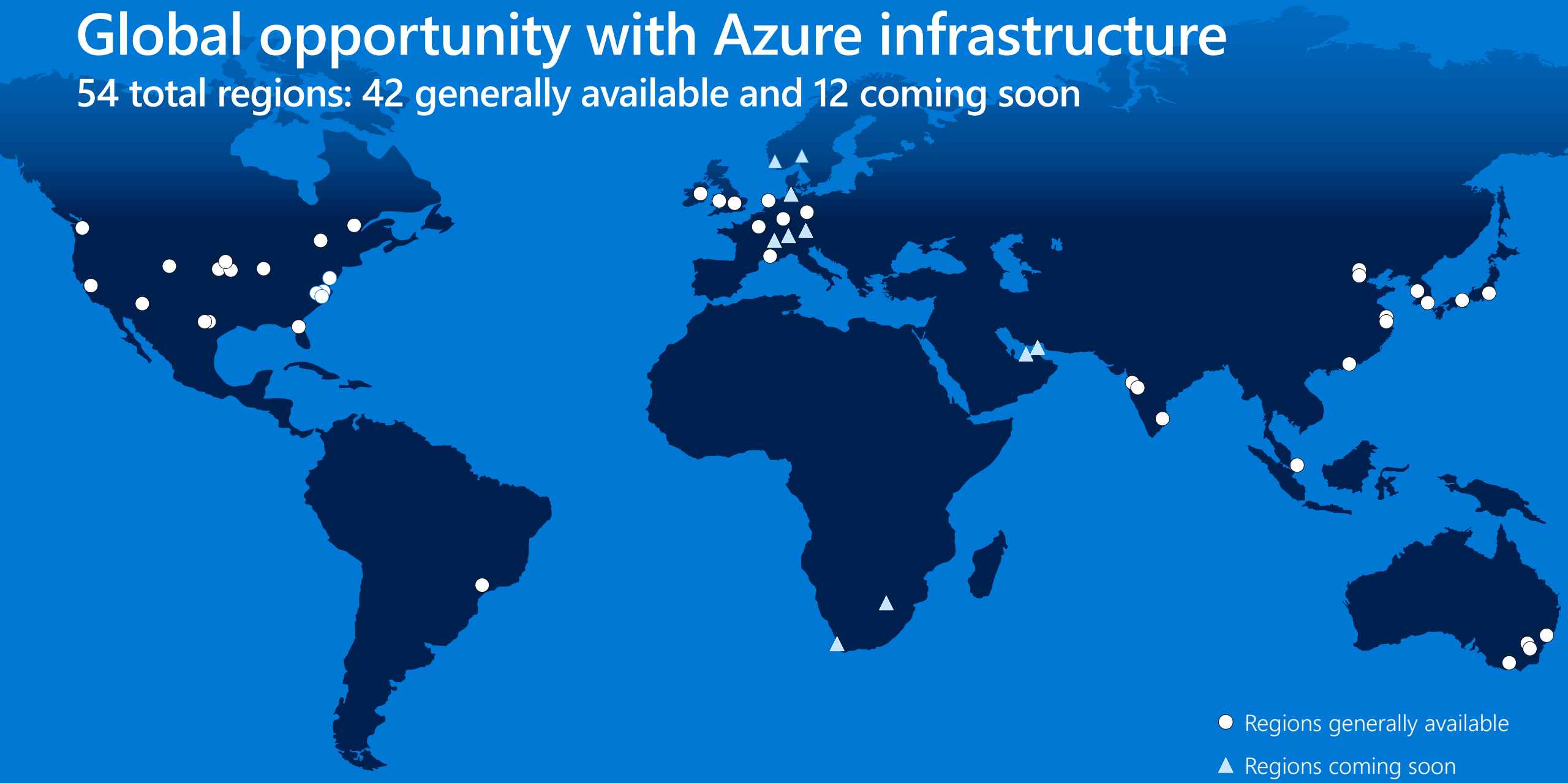
Azure solutions



Security and compliance

Global opportunity with Azure infrastructure

54 total regions: 42 generally available and 12 coming soon

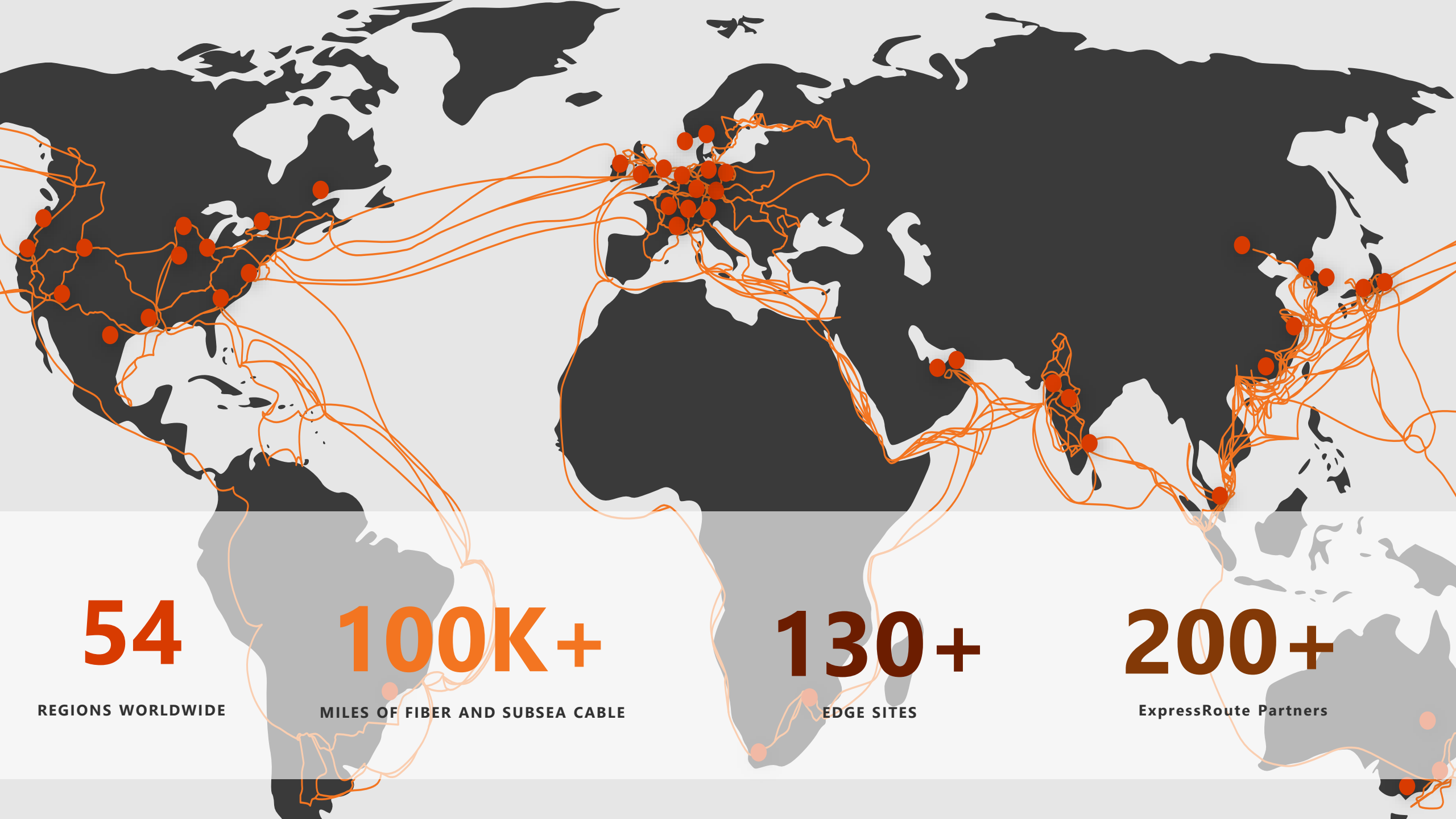


Quincy, WA




Cheyenne, WY


Dublin, Ireland



54

REGIONS WORLDWIDE

100K+

MILES OF FIBER AND SUBSEA CABLE

130+

EDGE SITES

200+

ExpressRoute Partners



One of the largest networks on the planet

Inter and intra regional connectivity

Terrestrial & subsea network

Interconnection

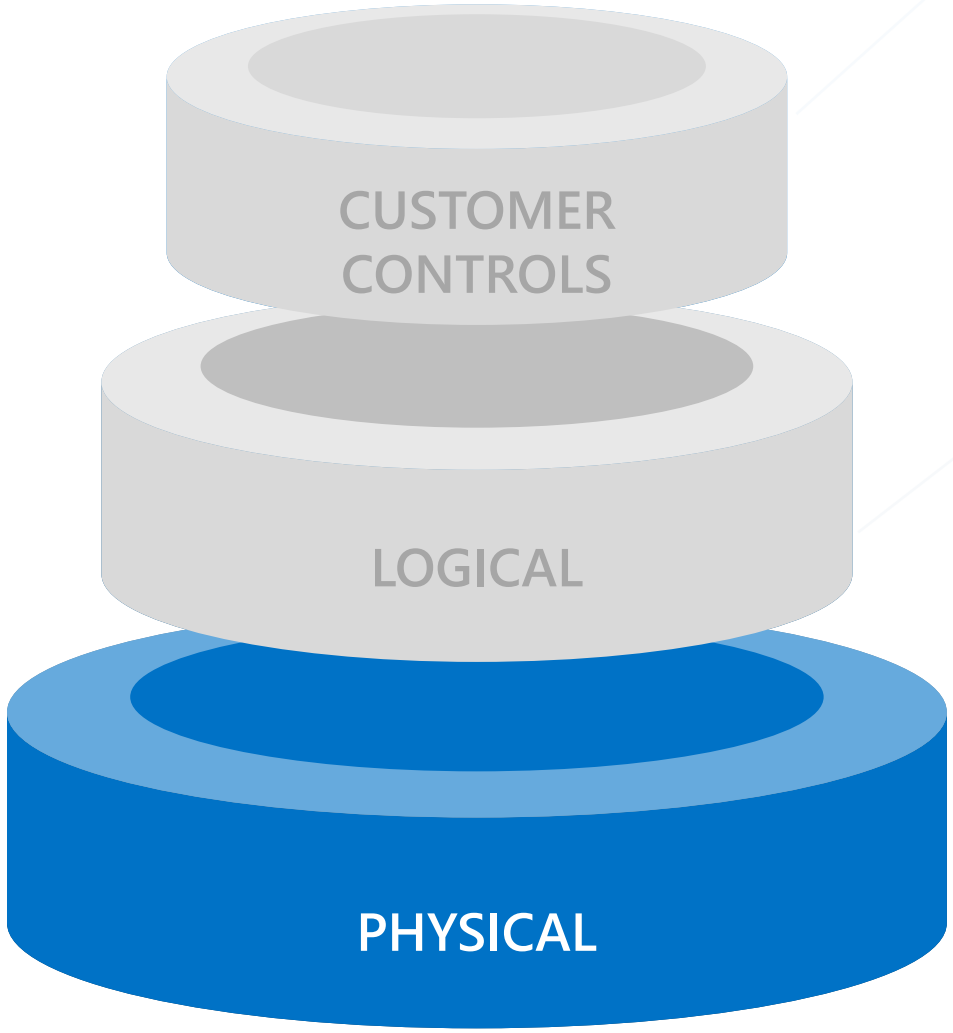
Terrestrial and Subsea Cable



Proactive investment enables agile expansion

Network innovation supports the growth of cloud services

Layered security approach



- Access approval
- Perimeter
- Building
- Server environment
- Data-bearing device controls

Defense in depth

Massive investment in physical security – people, process, technology

Access approval

Perimeter

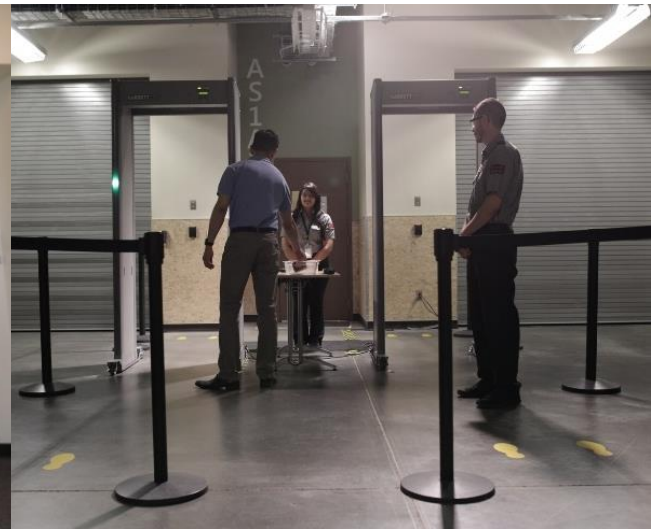
Building

Server environment









The most trusted and compliant cloud

Global

- ISO 27001
- ISO 27018
- ISO 27017
- ISO 22301
- SOC 1 Type 2
- SOC 2 Type 2
- SOC 3
- CSA STAR Self-Assessment
- CSA STAR Certification
- CSA STAR Attestation

US gov

- Moderate JAB P-ATO
- High JAB P-ATO
- DoD DISA SRG Level 2
- DoD DISA SRG Level 4
- SP 800-171
- FIPS 140-2
- Section 508 VPAT
- ITAR
- CJIS
- IRS 1075

Industry

- PCI DSS Level 1
- CDSA
- MPAA
- FACT UK
- Shared Assessments
- FISC Japan
- HIPAA /HITECH Act
- HITRUST
- GxP 21 CFR Part 11
- MARS-E
- IG Toolkit UK
- FERPA
- GLBA
- FFIEC

Regional

- Argentina PDPA
- EU Model Clauses
- UK G-Cloud
- China DJCP
- China GB 18030
- China TRUCS
- Singapore MTCS
- Australia IRAP/CCSL
- New Zealand GCIO
- Japan My Number Act
- ENISA IAF
- Japan CS Mark Gold
- Spain ENS
- Spain DPA
- India MeitY
- Canada Privacy Laws
- Privacy Shield
- Germany IT Grundschutz workbook

Azure covers 85 compliance offerings

The deepest and most comprehensive compliance coverage in the industry

Global

- ☑ ISO 27001:2013
- ☑ ISO 27017:2015
- ☑ ISO 27018:2014
- ☑ ISO 22301:2012
- ☑ ISO 9001:2015
- ☑ ISO 20000-1:2011
- ☑ SOC 1 Type 2
- ☑ SOC 2 Type 2
- ☑ SOC 3
- ☑ CIS Benchmark
- ☑ CSA STAR Certification
- ☑ CSA STAR Attestation
- ☑ CSA STAR Self-Assessment
- ☑ WCAG 2.0 (ISO 40500:2012)

US Gov

- ☑ FedRAMP High
- ☑ FedRAMP Moderate
- ☑ EAR
- ☑ ITAR
- ☑ DoD DISA SRG Level 5
- ☑ DoD DISA SRG Level 4
- ☑ DoD DISA SRG Level 2
- ☑ DFARS
- ☑ DoE 10 CFR Part 810
- ☑ NIST SP 800-171
- ☑ NIST CSF
- ☑ Section 508 VPATs
- ☑ FIPS 140-2
- ☑ CJIS
- ☑ IRS 1075

Industry

- ☑ PCI DSS Level 1
- ☑ GLBA (US)
- ☑ FFIEC (US)
- ☑ Shared Assessments (US)
- ☑ SEC 17a-4 (US)
- ☑ CFTC 1.31 (US)
- ☑ FINRA 4511 (US)
- ☑ SOX (US)
- ☑ 23 NYCRR 500 (US)
- ☑ OSFI (Canada)
- ☑ FCA + PRA (UK)
- ☑ APRA (Australia)
- ☑ FINMA (Switzerland)
- ☑ FSA (Denmark)
- ☑ RBI + IRDAI (India)
- ☑ MAS + ABS (Singapore)
- ☑ NBB + FSMA (Belgium)
- ☑ AFM + DNB (Netherlands)
- ☑ European Banking Authority (EBA)
- ☑ FISC (Japan)
- ☑ HIPAA BAA (US)
- ☑ HITRUST Certification
- ☑ GxP (FDA 21 CFR Part 11)
- ☑ MARS-E (US)
- ☑ NHS IG Toolkit (UK)
- ☑ NEN 7510:2011 (Netherlands)
- ☑ FERPA (US)
- ☑ CDSA
- ☑ MPAA (US)
- ☑ FACT (UK)
- ☑ DPP (UK)

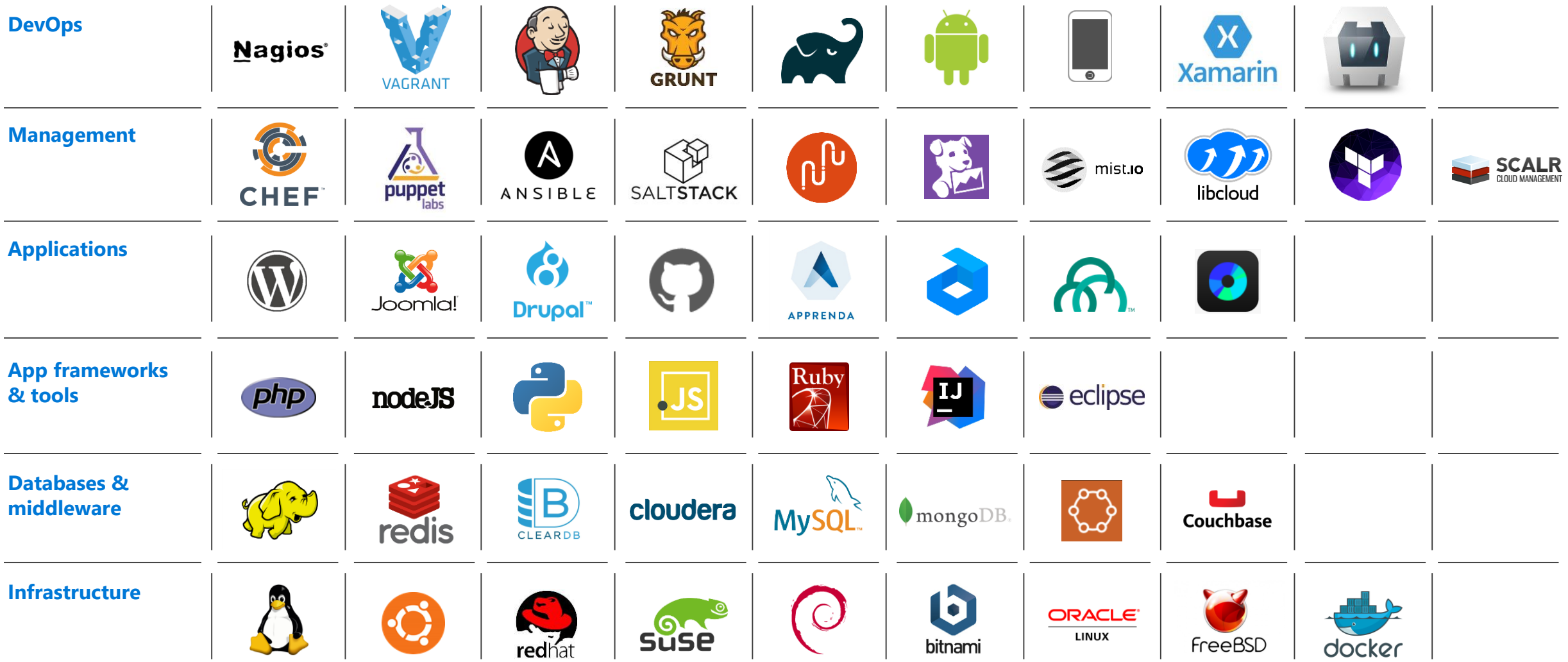
Regional

- ☑ Argentina PDPA
- ☑ Australia IRAP Unclassified
- ☑ Australia IRAP PROTECTED
- ☑ Canada Privacy Laws
- ☑ China GB 18030:2005
- ☑ China DJCP (MLPS) Level 3
- ☑ China TRUCS / CCCPPF
- ☑ EU EN 301 549
- ☑ EU ENISA IAF
- ☑ EU Model Clauses
- ☑ EU – US Privacy Shield
- ☑ GDPR
- ☑ Germany C5
- ☑ Germany IT-Grundschutz workbook
- ☑ India MeitY
- ☑ Japan CS Mark Gold
- ☑ Japan My Number Act
- ☑ Netherlands BIR 2012
- ☑ New Zealand Gov CIO Framework
- ☑ Singapore MTCS Level 3
- ☑ Spain ENS High
- ☑ Spain DPA
- ☑ UK Cyber Essentials Plus
- ☑ UK G-Cloud
- ☑ UK PASF

https://aka.ms/AzureCompliance

Tools

Build and innovate with choice and flexibility



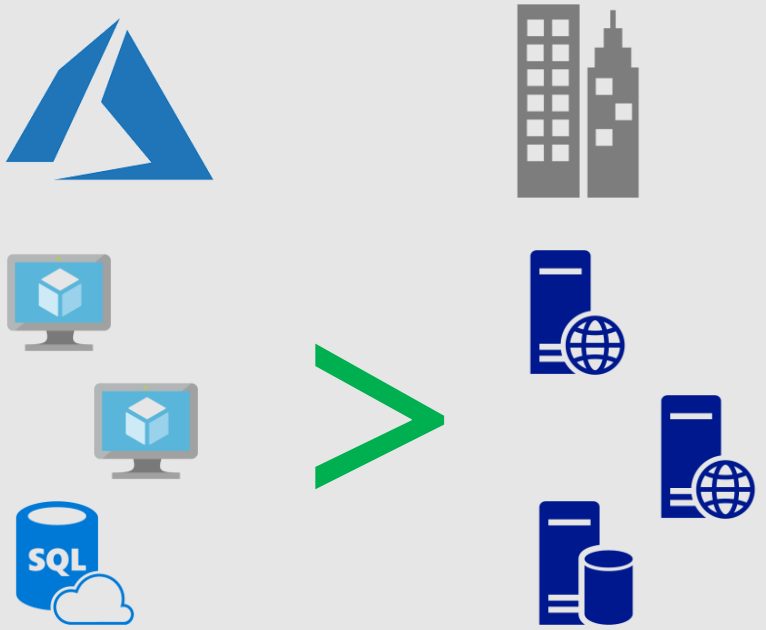
Core infrastructure

Drive cost and operational efficiencies

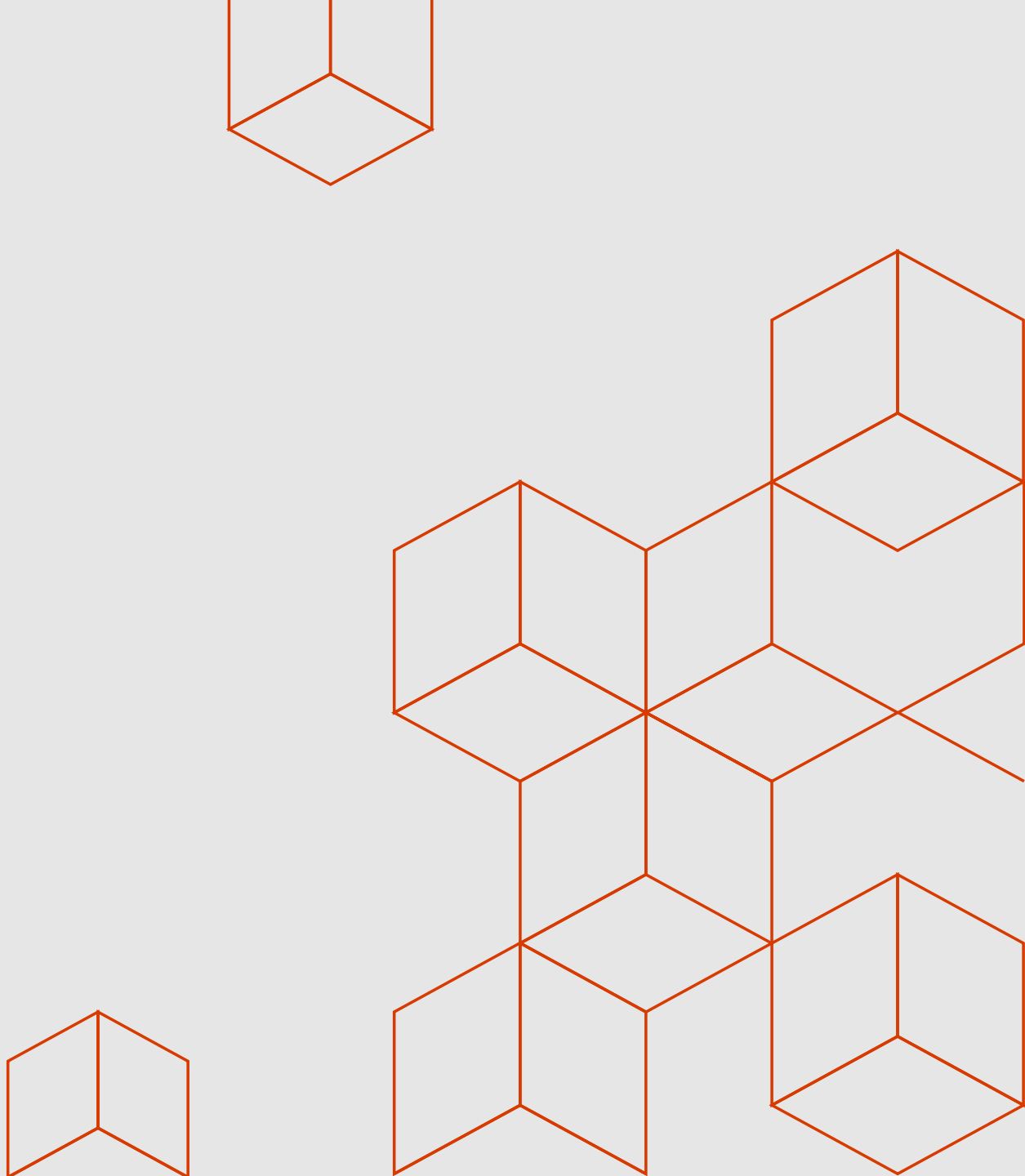
Azure IaaS offers vastly improved capabilities in optimization and agility capabilities than todays on-premises environments

Take advantage of the benefits

- Smaller gap in existing skills
- Microsoft manages the hardware
- Only pay for what you use
- Automated provisioning and de-provisioning of infrastructure
- Management and monitoring efficiency



Azure Compute

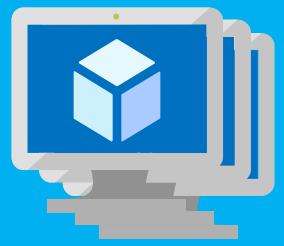


Virtual Machines & Scale Sets



Virtual Machines

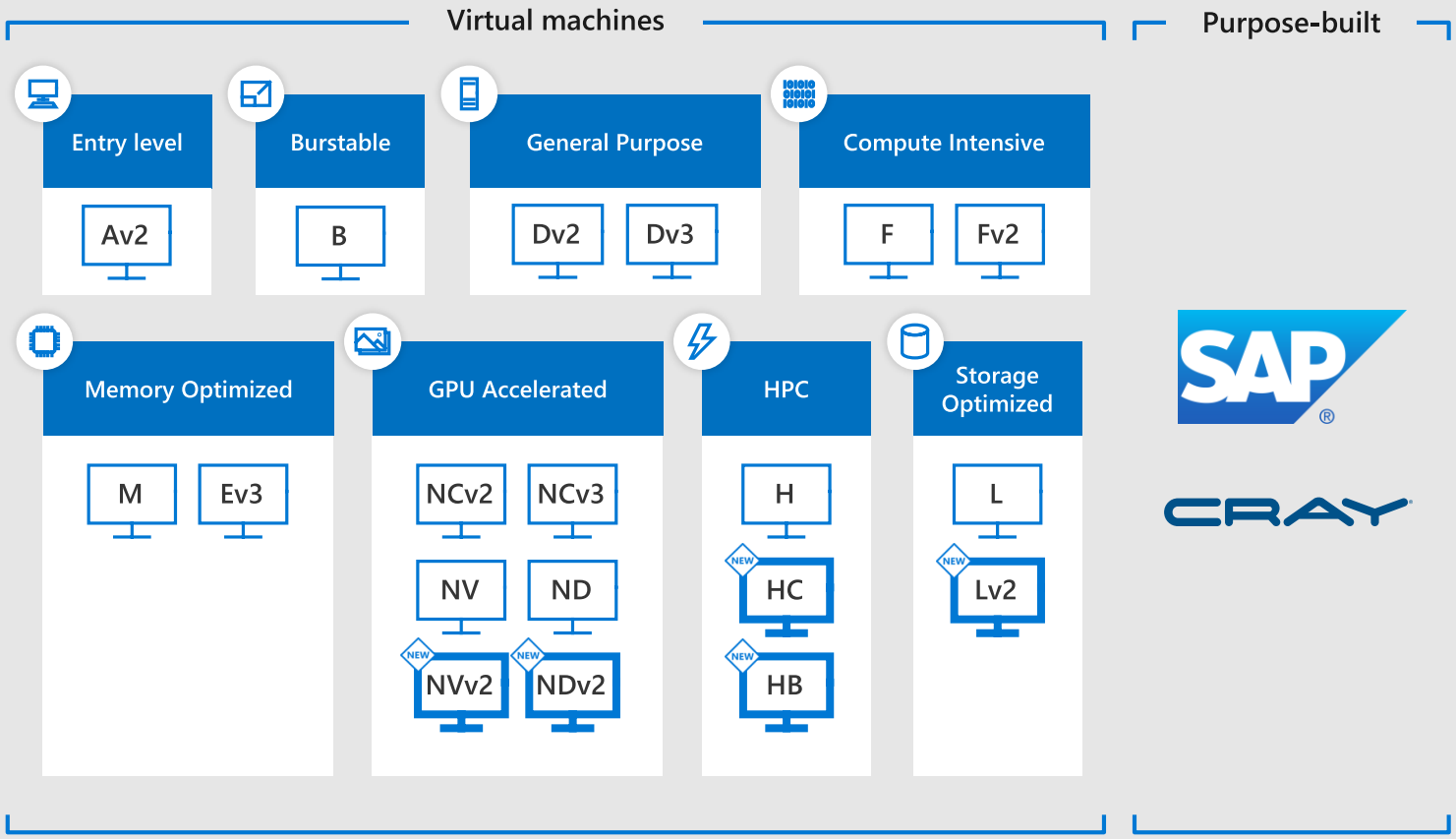
- Minimal learning curve for existing virtualization pros
- Run most Windows or Linux-based workloads
- Easiest target for migrating to the cloud
- Flexible instance sizes – use the best fit for your workload
- Advanced networking and storage capabilities



Virtual Machine Scale Sets

- All of the power and flexibility of a Virtual Machine
- Designed for high-end workloads
- Automatically scale up (or down) to 1000 instances

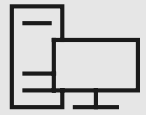
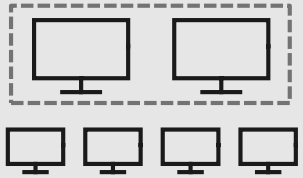
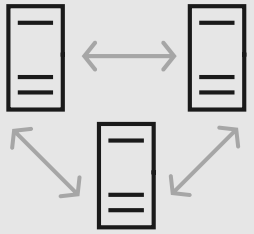
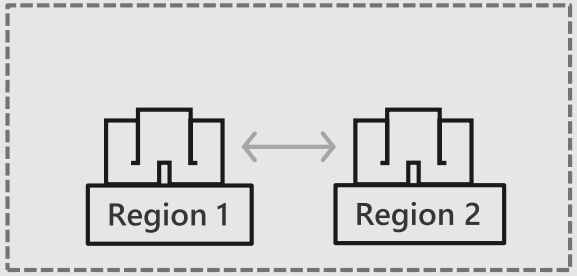
Compute options for *every* workload



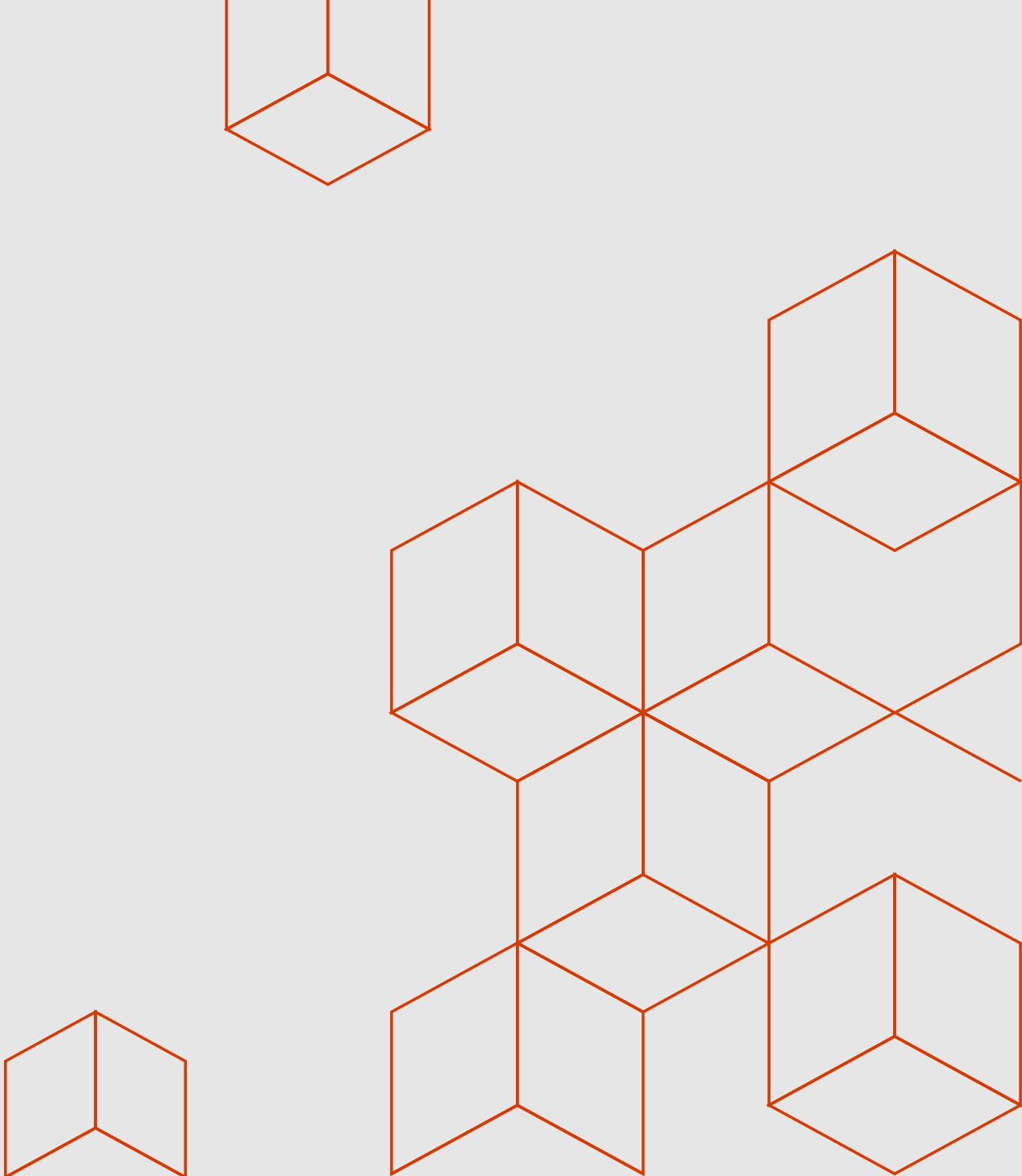
What's new and on the roadmap?

- NVv2 for graphic intensive applications in Preview
- NDv2 accelerate machine training and HPC New Ev3 and Esv3
- Lv2 VMs currently in limited public preview.
- H-series VMs coming soon:
- HC for computationally intensive workloads
- HB for memory bandwidth driven workloads

High availability options

Industry-only	High availability SLA		Disaster recovery
VM SLA 99.9%	**VM SLA** 99.95%	**VM SLA** 99.99%	**Regions** 54
			
Single VM Protection with Premium Storage	**Availability sets** Protection against failures within datacenters	**Availability zones** Protection from entire datacenter failures	**Region pairs** Protection from disaster with Data Residency compliance

Storage



Azure Storage Options

Disks
Persistent disks for Azure IaaS VMs

Premium Storage Disks option: SSD based, high IOPS, low latency

Files
Fully Managed File Shares in the Cloud

SMB and REST access

"Lift and shift" legacy apps

Blobs
Highly scalable, REST based cloud object store

Block Blobs: Sequential file I/O
Cool Tier Available
Page Blobs: Random-write pattern data
Append Blobs

Hybrid
Azure File Sync

StorSimple

Transport
Storage Explorer

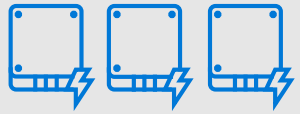
Command Line

Import / Export

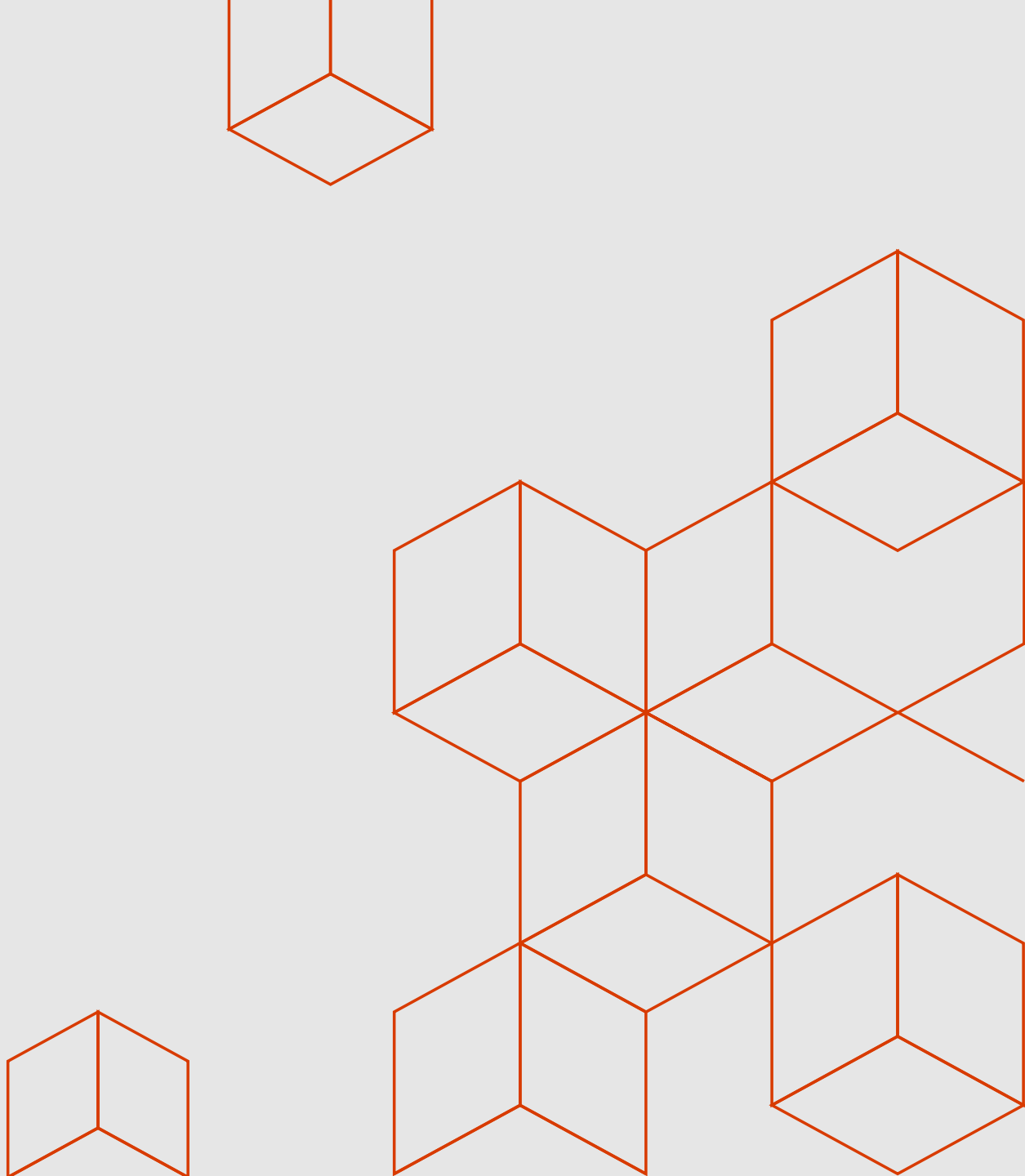
Data Box

Built on a unified Distributed Storage System
Durability, Encryption at Rest, Strongly Consistent Replication, Fault Tolerance, Auto Load-Balancing

Virtual Machine Disk Storage

	Standard HDD	Standard SSD (Now GA)	Premium SSD	Ultra SSD (Preview (NEW))
				
Single disk max value	Low-cost storage	Consistent performance	High performance	Sub-millisecond latency
SIZE	32TiB	32TiB	32TiB	64TiB
IOPS	2,000	2,000	20,000	80,000 – 160,000
BANDWIDTH	500 MBps	500 MBps	750 MBps	2,000 MBps

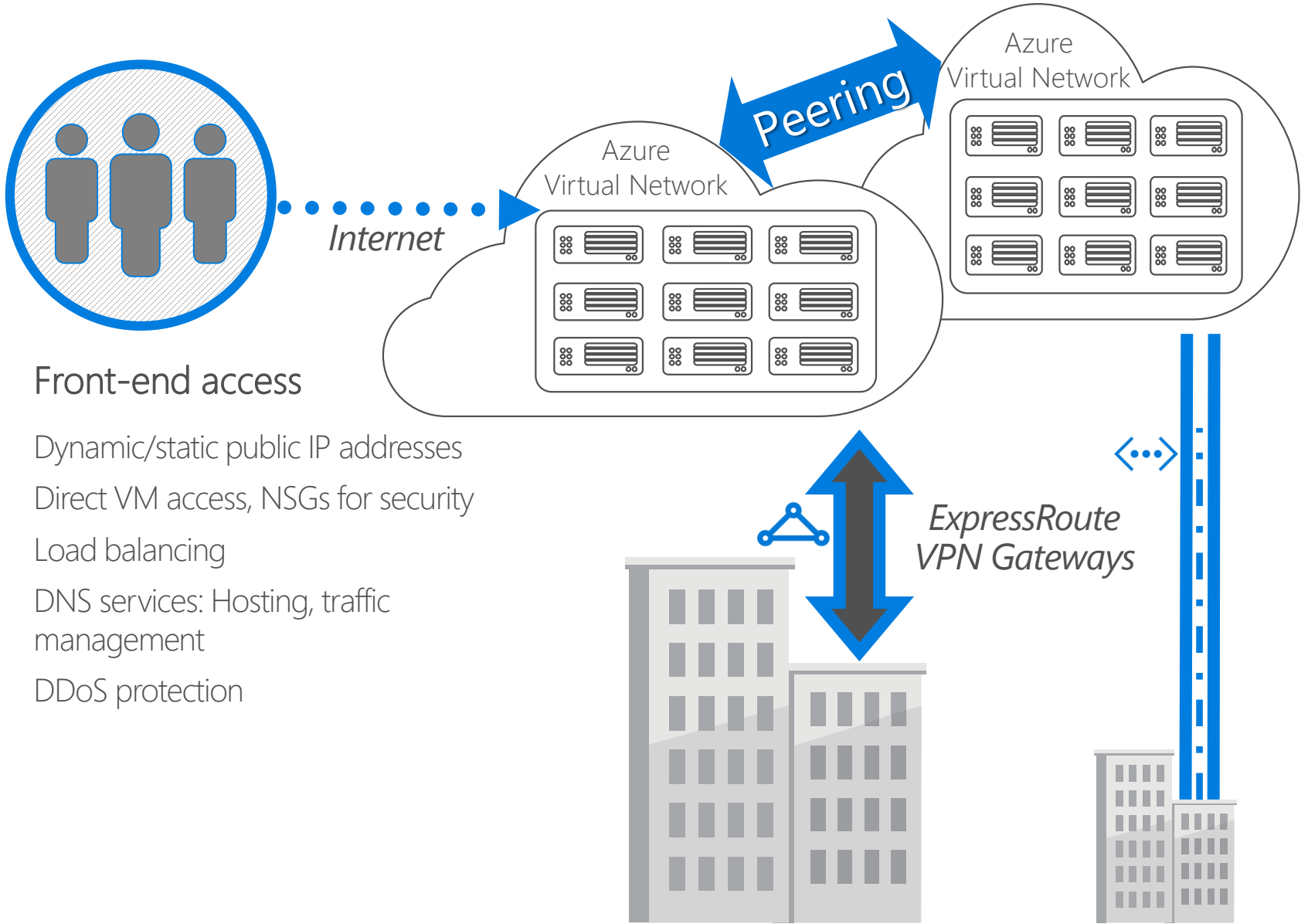
Azure Networking



Hybrid Connectivity

	Secure point-to-site connectivity		• POC Efforts • Small scale deployments • Connect from anywhere
	Secure site-to-site VPN connectivity		• Connect to Azure compute from on-premises or another Azure region
	VNet Peering within region		• In-region VNet-to-VNet connectivity • Direct VM-to-VM connectivity • Peer VNets for routing and transit
	ExpressRoute private connectivity		• Private connectivity from your on-premises data center to Azure virtual networks and PaaS Services

The big (network) picture



Front-end access

Dynamic/static public IP addresses

Direct VM access, NSGs for security

Load balancing

DNS services: Hosting, traffic management

DDoS protection

Virtual Network

"Bring Your Own Network"

Segment with subnets and security groups

Control traffic flow with user-defined routes

VNet peering

Back-end connectivity

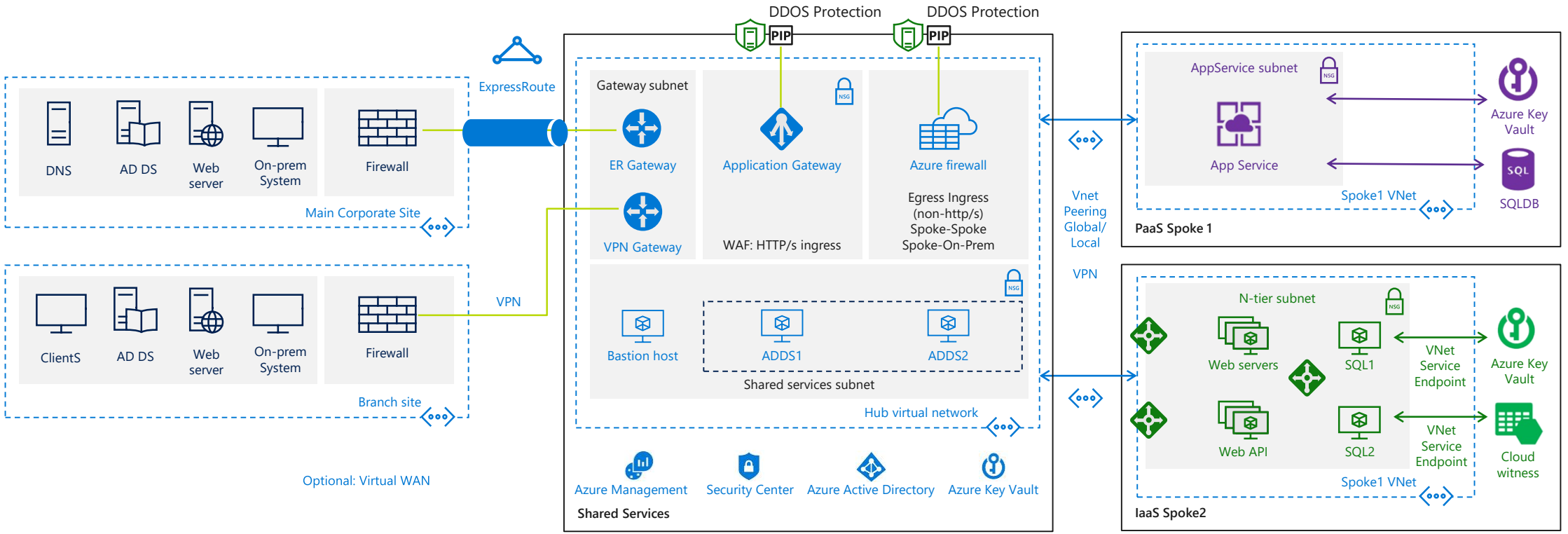
Point-to-site for dev/test

Peering for cross –VNet connectivity in the same region

VPN Gateways for secure cross region connectivity

ExpressRoute for private enterprise grade connectivity

Sample Network Architecture



Connect

- Virtual Network
- Virtual WAN
- VPN : Site-to-site / Point-to-Site
- ExpressRoute
- DNS, User Defined Routes

Protect

- DDoS Protection
- Firewall / Web Application Firewall
- Network Security Group
- Virtual Network Endpoints
- Virtual Appliances

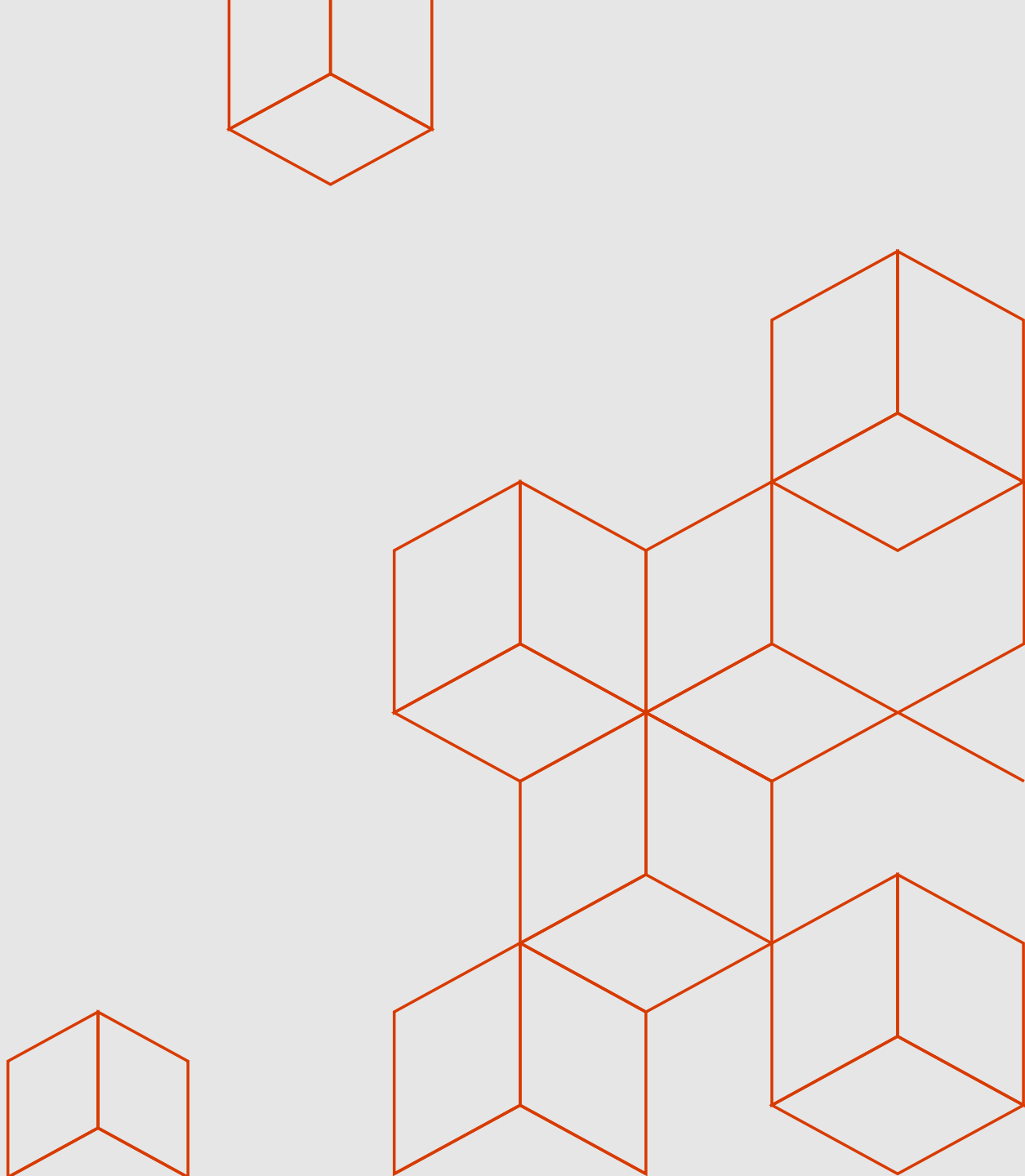
Deliver

- CDN
- Front Door
- Traffic Manager
- Application Gateway
- Load Balancer

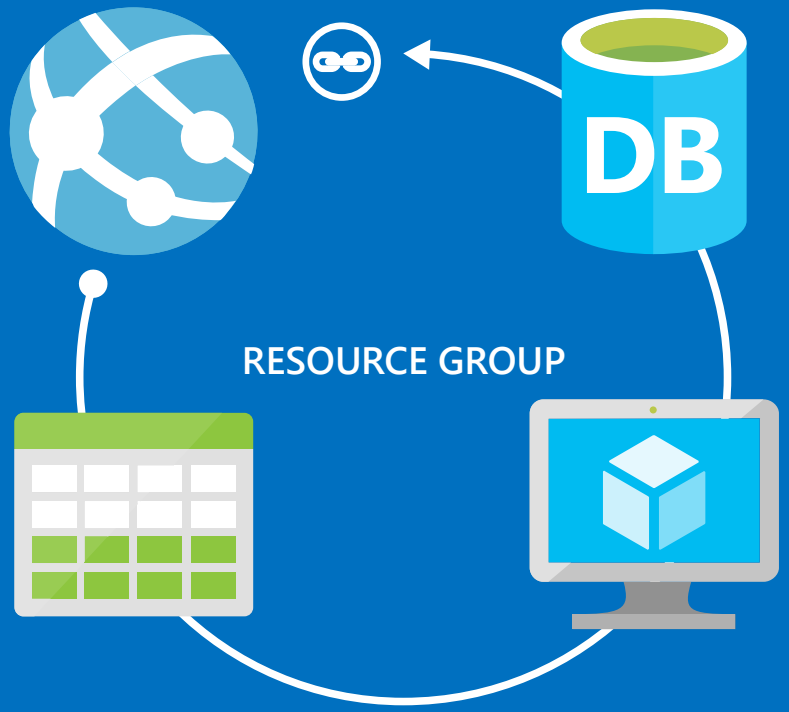
Monitor

- Network Watcher
- ExpressRoute Monitor
- Azure Monitor
- Virtual Network TAP

Management



Azure Resource Manager



Enable application management within Azure

Resource groups are containers that can contain multiple IaaS + PaaS resources

Templatize application deployment and configuration, supports DevOps

Governance and control with Role Based Access Control (RBAC), policies and resource locks

Enterprise Grade Cloud Management

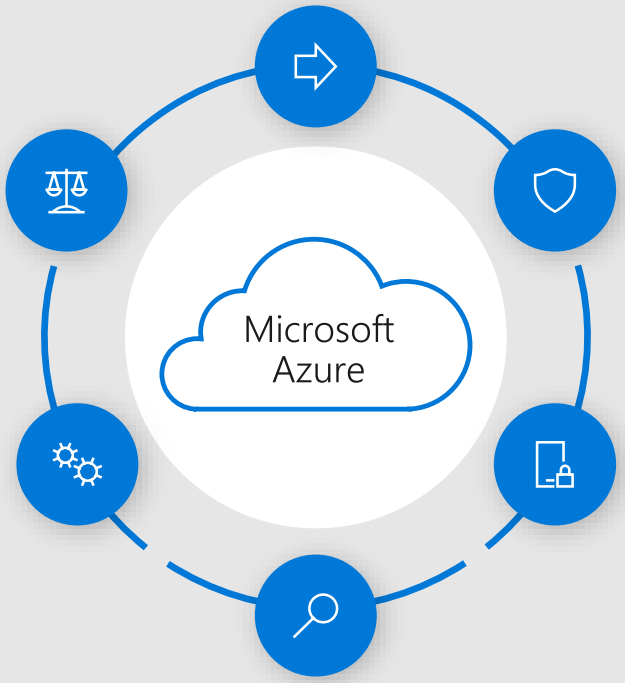


Operations

- Auto-shutdown
- Backup
- Disaster recovery
- Update management
- Inventory
- Change tracking
- Configuration management ...
- Run command

Govern
Identity & Access Management
Policy Based Management
Cost Management

Configure
Configuration
Update Management
Automation
Scripting



Migrate

Secure
Security Management
Threat Protection

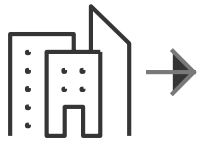
Protect
Backup
Disaster Recovery

Monitor
App, Infra, and Network Monitoring
Log Analytics & Diagnostics

Simplified VM Management to Comprehensive Cloud Services

Migration

Summarizing cloud migration strategies



	Rehost (lift & shift)	Refactor	Rearchitect	Rebuild
Description	**Redeploy as-is to cloud**	**Minimally alter to take better advantage of cloud**	**Materially alter/decompose application to services**	**New code written with cloud native approach**
Drivers	• Reduce Capex • Free up datacenter space • Quick cloud ROI	• Faster, shorter, updates • Code portability • Greater cloud efficiency (resources, speed, cost)	• App scale and agility • Easier adoption of new cloud capabilities • Mix technology stacks	• Accelerate innovation • Build apps faster • Reduce operational cost
Technologies	**IaaS**	**Containers** **PaaS**	**PaaS** **Serverless** **Microservices**	

Choice of tools for every stage and every requirement

Microsoft

Assess

Azure Migrate

Data Migration Assistant

SQL Server Migration Assistant

Migrate

Azure Site Recovery

Azure Database Migration Service

Azure Data Box

Optimize

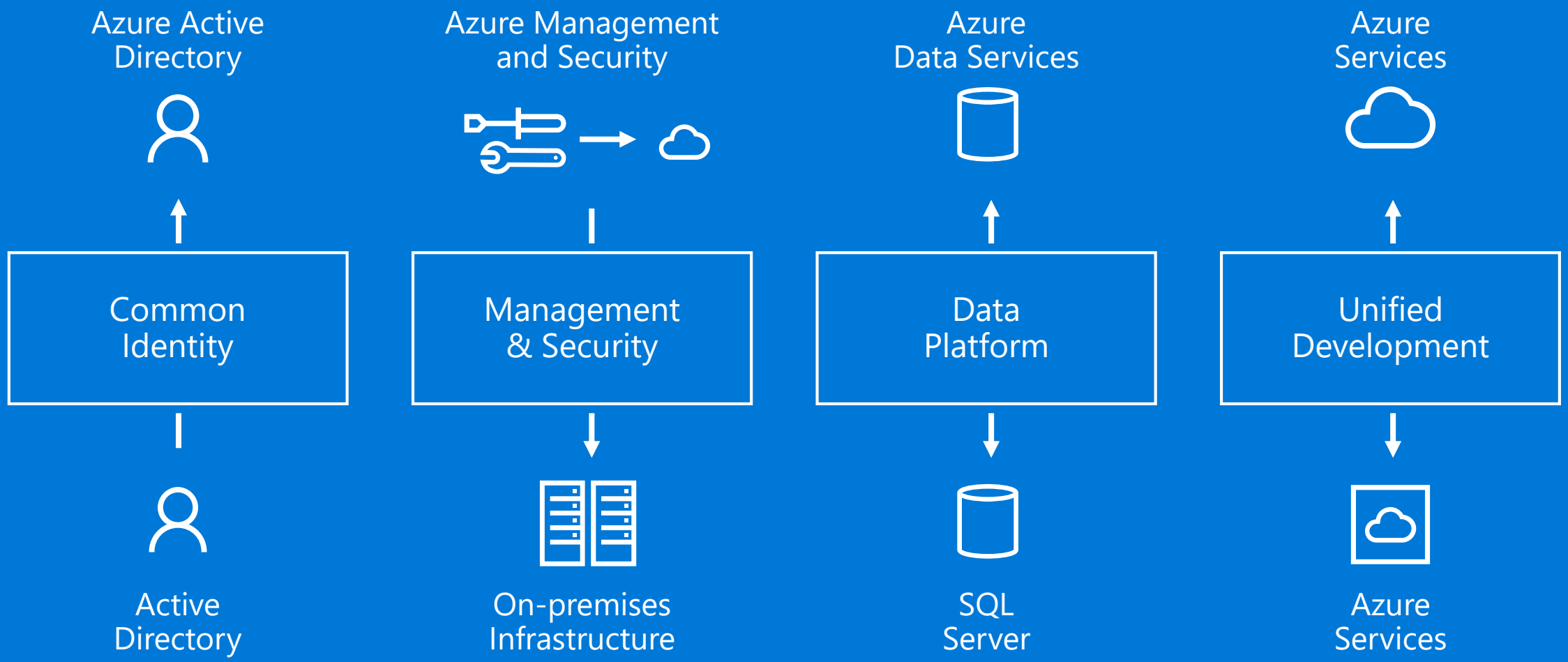
Azure security and management (security, backup, monitoring, cost management)

Partners

MOVERE
Cloudamize
CloudAtlas®
BitTitan
STRATOZONE®
turbonomic
CloudEndure®
Ispirer
Corent
intigua
stiim
ATTUNITY
DATOMETRY
MORPHIS
Informatica
QuerySurge

Azure Stack + hybrid

Overcome complexity with consistent, hybrid cloud



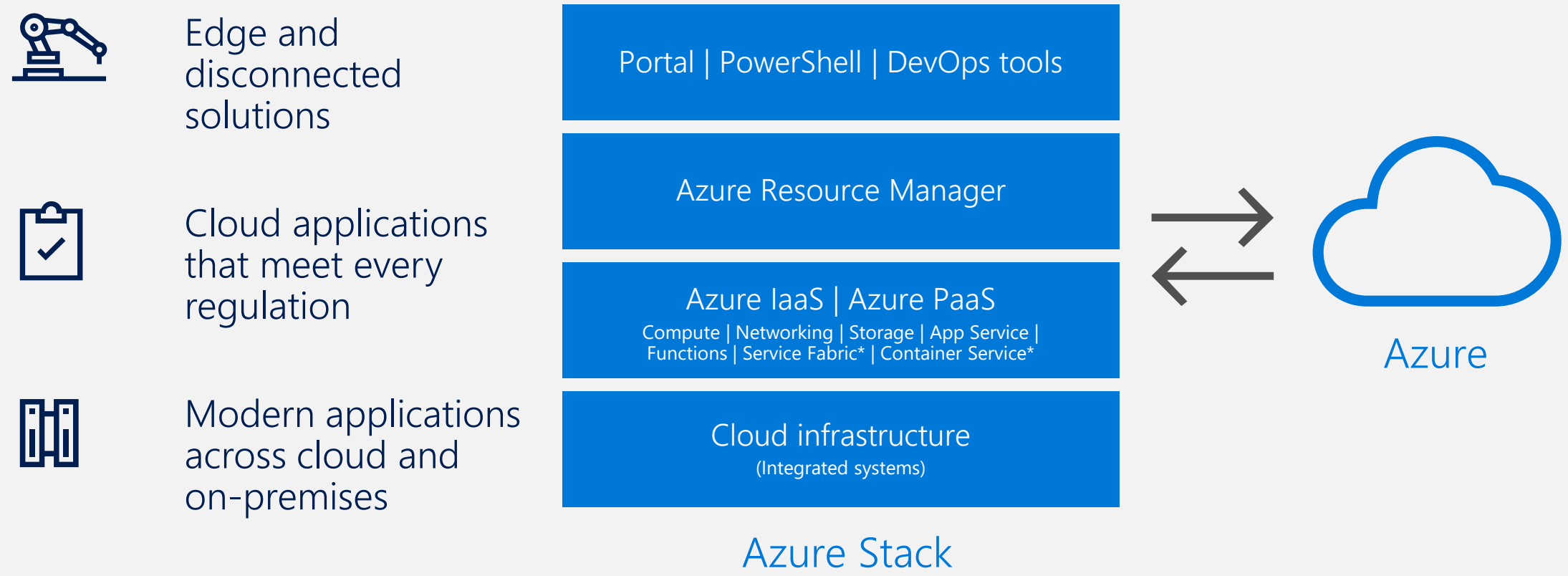
Truly consistent hybrid cloud platform

Bring the agility and innovation of the cloud to your datacenter

Edge and disconnected solutions

Cloud applications that meet every regulation

Modern applications across cloud and on-premises



Advanced workloads

Microsoft Azure

TOOLS

Visual Studio + Visual Studio Code + Visual Studio Team Services (DevOps)

ADVANCED WORKLOADS

Web + Mobile

Microservices & Container

Serverless

Workflow & Integration

API Management

Identity Management

Data + Analytics

Cognitive Services

High Performance Compute

Internet of Things

CORE INFRASTRUCTURE

Compute Storage Networking Security

Azure Stack + Hybrid

Choice + Flexibility

DevOps



Applications



App Frameworks



Databases & Middleware



Infrastructure



Azure is an open cloud



Build with a platform designed for future growth



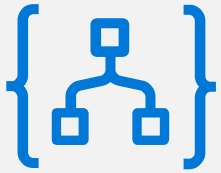
App service

Web, mobile, and API apps



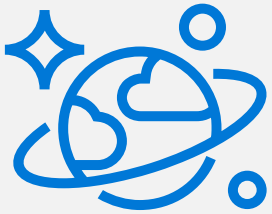
DevTest Labs

Integrate with your existing toolchain



Logic apps

Workflow & integration



Cosmos DB

Globally distributed, multi-model database service

RECENT ANNOUNCEMENTS

Data Lake Storage and Analytics

Azure Container Instances service

Azure Container Service (AKS)

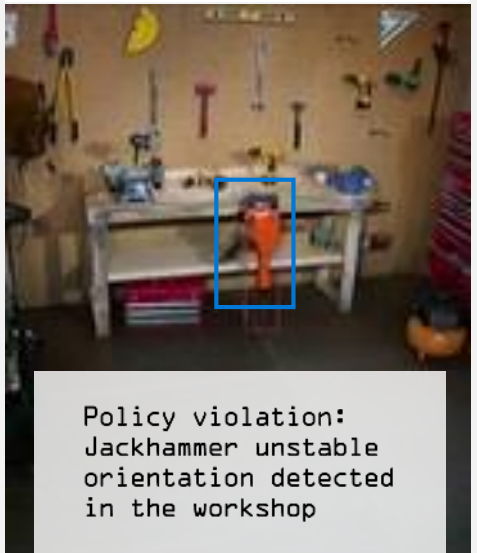
Add a natural, human touch with Cognitive Services

Vision

Computer Vision

Listening...

Object recognition...

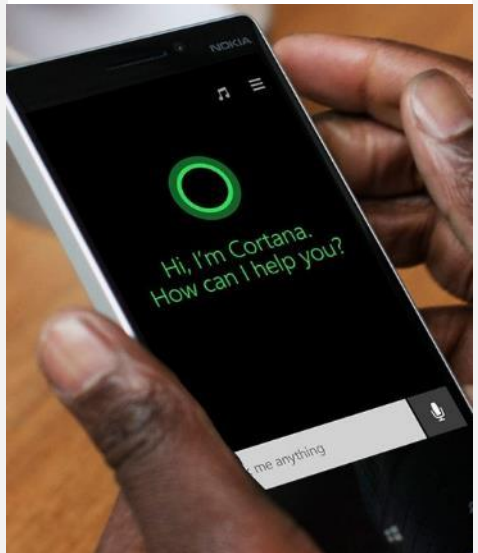


Speech

Speech-to-text and text-to-speech service

Listening...

Convert this text to speech please...

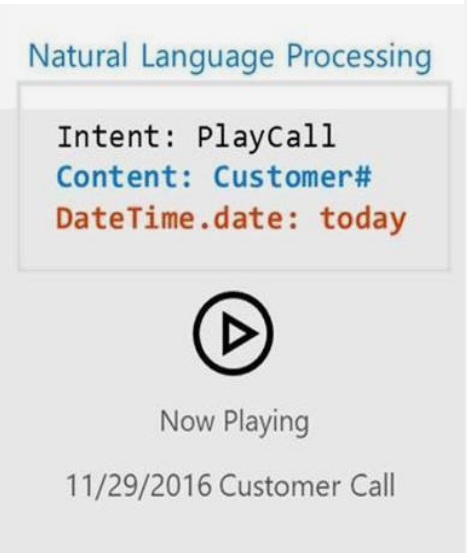


Language

Natural Language Processing

Listening...

Play today's conference call...

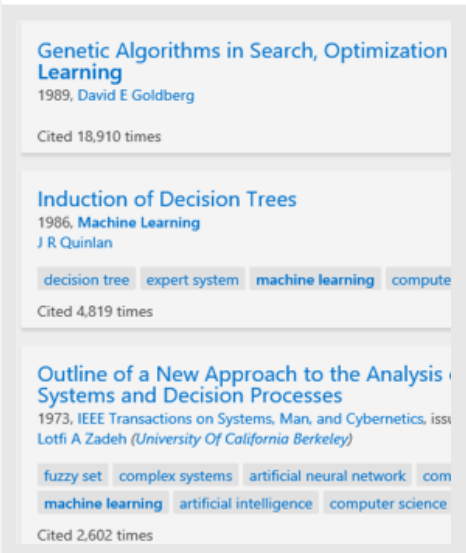


Knowledge

Knowledge Exploration Service

Listening...

Top publications in AI...

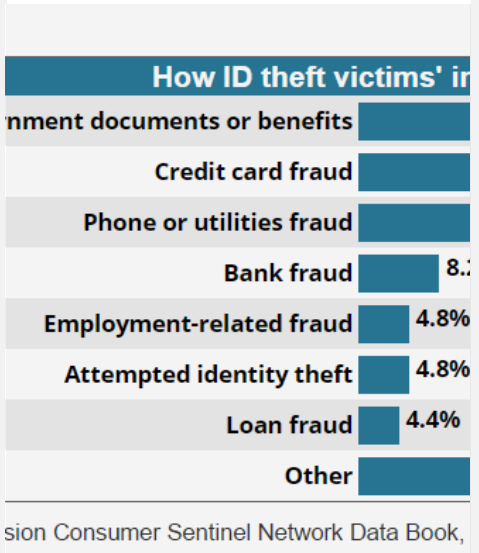


Search

Bing Search

Listening...

Fraud prevention results...



Simplify IoT with preconfigured solutions



Azure IoT Suite

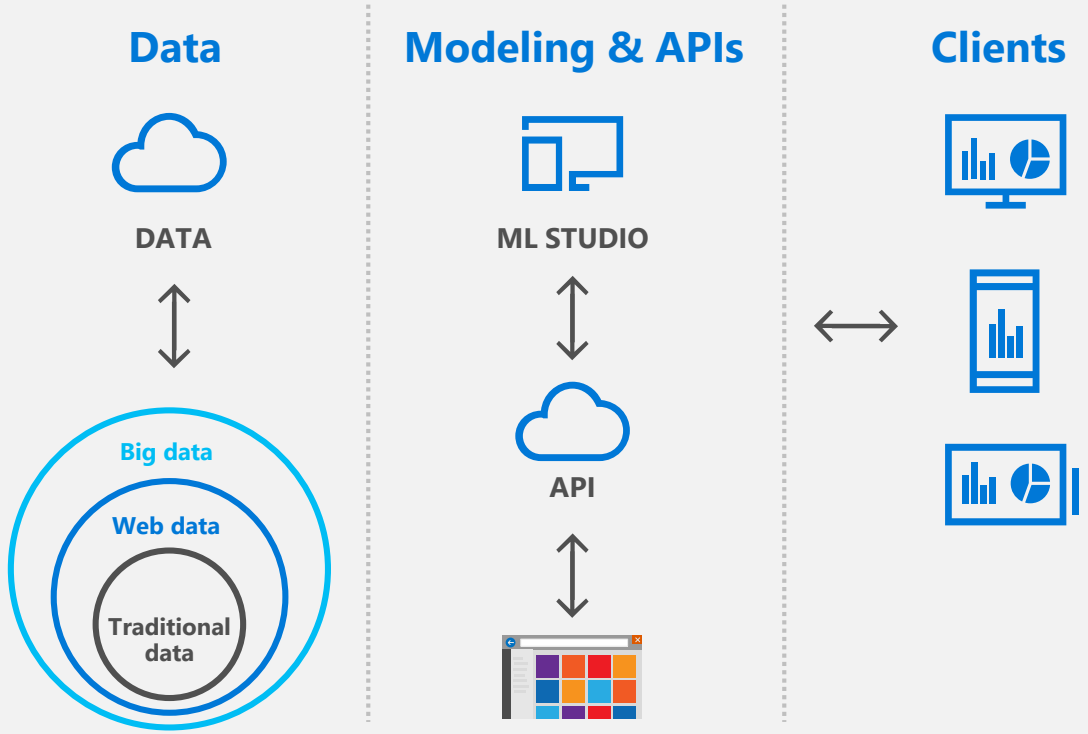
- Device connectivity & management
- Data ingestion and command & control
- Stream processing & predictive analytics
- Workflow automation and integration
- Dashboards and visualization
- Preconfigured solutions
  - Remote monitoring
  - Predictive maintenance
  - Connected factory (new)

Predict customer needs with powerful tools

Azure Machine Learning templates

- Demand forecasting
- Predictive maintenance
- Vehicle telemetry

Azure Machine Learning services




Thank You!
ευχαριστώ
Salamat Po
متشكرم
شكراً
Grazie
благодаря
ありがとうございます
Kiitos
Teşekkürler
谢谢
ขอบคุณครับ
Obrigado
شكريه
Terima Kasih
Dziękuję
Hvala
Köszönöm
Tak
Dank u Wel
дякую
Tack
Mulţumesc
спасибо
Danke
Cám ơn
Gracias
多謝晒
Ďakujem
תודה
நன்றி
Děkuji
감사합니다